



Clinical and radiographic findings of gingival and periodontal diseases



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Learning outcomes

- **Demonstrate the clinical and radiographic features of gingival diseases**
- **Demonstrate the clinical and radiographic features of periodontal diseases**





Clinical features of gingivitis

- + **Plaque** is the primary etiologic factor that causes gingivitis
 - + Clinical features of gingivitis can be any of the following clinical signs: *redness and sponginess of the gingival tissue, bleeding on provocation, changes in contour, and the presence of calculus or plaque*
 - + **No radiographic evidence of crestal bone loss**
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Description

+ *Localized gingivitis*-- confined to the gingiva of one tooth or group of teeth

+ *Generalized gingivitis* -- involves the entire mouth

+ *Marginal gingivitis*-- involves the gingival margin, and can include a portion of the contiguous attached gingiva

+ *Papillary gingivitis*-- involves the interdental papillae, and it often extends into the adjacent portion of the gingival margin

+ *Diffuse gingivitis* -- affects the gingival margin, the attached gingiva, and the interdental papillae

Earliest signs of gingivitis often occur in the interdental papillae

Plaque and calculus

- + **Dental plaque** is defined clinically as a structured, resilient, yellow-grayish substance that adheres tenaciously to the intraoral hard surfaces, including removable and fixed restorations
- + **Calculus** consists of mineralized bacterial plaque that forms on the surfaces of natural teeth and dental prostheses and is usually covered by a layer of unmineralized plaque



Gingival Bleeding on Probing (BOP)

- + **Two earliest signs of inflammation-- increased gingival crevicular fluid production and bleeding on gentle probing**
- + **Varies in severity, duration, and ease of provocation**
- + **Easily detected clinically--value for early diagnosis**
- + **Appears earlier than a change in color or other visual signs of inflammation**
- + **An objective sign that requires less subjective estimation by the examiner**



Causes of gingival bleeding

- + **Most common cause of chronic or recurrent bleeding is chronic inflammation and BOP is provoked by mechanical trauma**
- + **Acute bleeding-- caused by injury, laceration by toothbrush bristles or by sharp pieces of hard food, gingival burns from hot foods or chemicals**
- + **Spontaneous bleeding -- necrotizing gingivitis**
- + **Systemic causes--vitamin C deficiency, platelet disorders, vitamin K deficiency, coagulation defects (e.g., hemophilia, leukemia, Christmas disease), etc**
- + **Medications--antiplatelet medications (e.g., aspirin) or anticoagulants (e.g., warfarin), oral contraceptives**



Color Changes in the Gingiva

- + Color is determined by the number and size of blood vessels, epithelial thickness, quantity of keratinization, and pigments in the epithelium
- + Red --when vascularization increases or the degree of epithelial keratinization is reduced or disappears
- + Pale --when vascularization is reduced or epithelial keratinization increases
- + Venous stasis contributes a bluish hue
- + Color changes vary with the intensity of inflammation





Changes in Gingival Consistency

- + Normal-- firm and resilient
- + In patients with chronic gingivitis, destructive (i.e., edematous) and reparative (i.e., fibrotic) changes coexist, and the consistency of the gingiva is determined by their relative predominance



Changes in Gingival Surface Texture

- ❖ Surface of normal gingiva exhibits numerous small depressions and elevations giving an orange-peel appearance referred to as *stippling*
- ❖ Stippling—seen in attached gingiva, but extends to various degrees into the interdental papilla
- ❖ Loss of stippling is an early sign of gingivitis
- ❖ In inflammation--smooth and shiny or firm and nodular
- ❖ Hyperkeratosis results in a leathery texture, and drug-induced gingival overgrowth produces a nodular surface

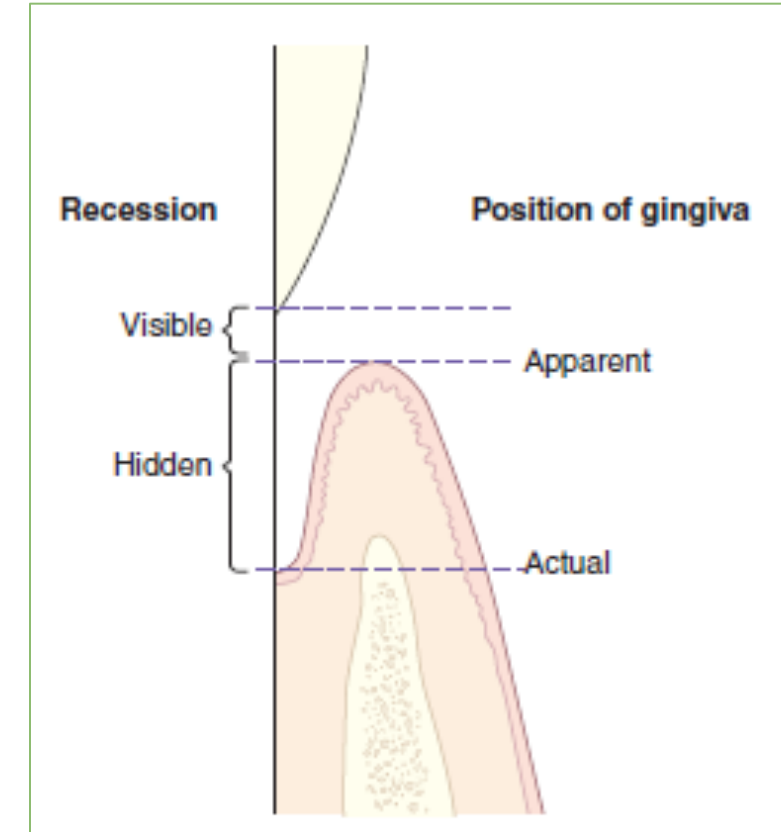
Stippling and Its Clinical Importance

- ✦ Stippling, is an external reflection of the underlying connective tissue projections into the overlying epithelium
- ✦ Presence of stippling is indicative of gingival health, and this surface feature is usually lost when the tissue is edematous



Changes in Gingival Position

- *Traumatic Lesions* — chemical, physical, or thermal--can result in gingival recession
- Gingival Recession--prevalence, extent, and severity increase with age, and more prevalent among males
- Definition of recession ---exposure of root surface by an apical shift in the position of the gingiva
- Actual position--level of the coronal end of the epithelial attachment on the tooth
- Apparent position--level of the crest of gingival margin



- + **Recession refers to location of gingiva rather than to its condition i.e. gingiva can be inflamed, or be normal except for its position**
- + **Localized or generalized**
- + **Causes of recession:**
 - faulty toothbrushing technique (i.e., gingival abrasion),*
 - tooth malposition*
 - friction from the soft tissues*
 - gingival inflammation*
 - abnormal frenum attachment*
 - iatrogenic dentistry*



Clinical Significance



- + **Exposed root surfaces are susceptible to caries**
- + **Abrasion or erosion of the cementum exposed by recession—sensitive**
- + **Hyperemia of the pulp -- from excessive exposure of the root surface**
- + **Interproximal recession creates oral hygiene problems results in plaque accumulation**
- + **Esthetic concern**



Changes in Gingival Contour

- Changes in gingival contour -- associated with gingival enlargement, but may also occur with other conditions
- **Stillman cleft** -- narrow, triangular shaped gingival recession
- **McCall festoon** -- rolled, thickened band of gingiva seen adjacent to cuspids as recession approaches MGJ
- Both represent peculiar inflammatory changes of the marginal gingiva



Stillman cleft



McCall festoon



Clinical features of periodontitis



- ✦ Clinical feature that distinguishes periodontitis from gingivitis -- clinical attachment loss as a result of inflammatory destruction of the periodontal ligament and alveolar bone
- ✦ This loss is often accompanied by periodontal pocket formation and changes in the density and height of the alveolar bone
- ✦ All types of periodontitis share histopathologic features, such as tissue changes in the periodontal pocket, mechanisms of tissue destruction, and healing mechanisms



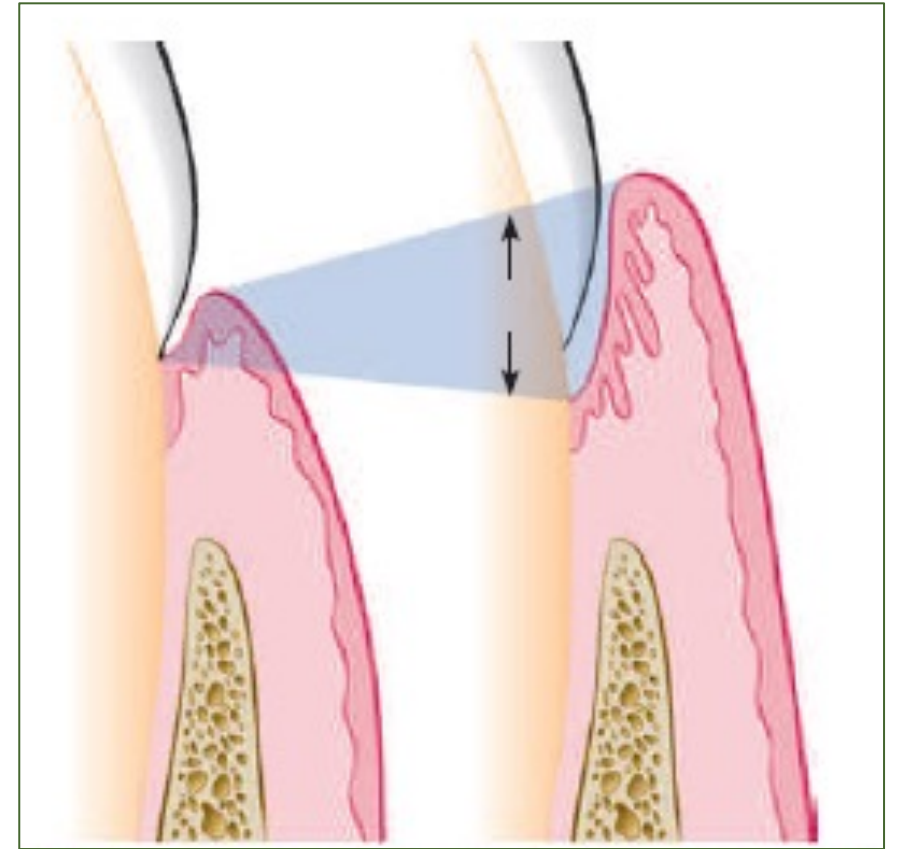
+ **Characteristic clinical findings in patients with untreated periodontitis include:**

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- 1. Supra and subgingival plaque and calculus***
 - 2. Gingival changes--swelling, redness, and loss of gingival stippling, altered margins***
 - 3. Altered gingival margins***
 - 4. Bleeding on probing***
 - 5. Pocket formation***
 - 6. Attachment loss***
 - 7. Bone loss (angular/vertical or horizontal)***
 - 8. Root furcation involvement***
 - 9. Increased tooth mobility***
 - 10. Change in tooth position***
 - 11. Tooth loss***



Pocket formation

- + The periodontal pocket, which is defined as a pathologically deepened gingival sulcus
- + Deepening of the gingival sulcus may occur as a result of coronal movement of the gingival margin, apical displacement of the gingival attachment, or a combination of the two processes

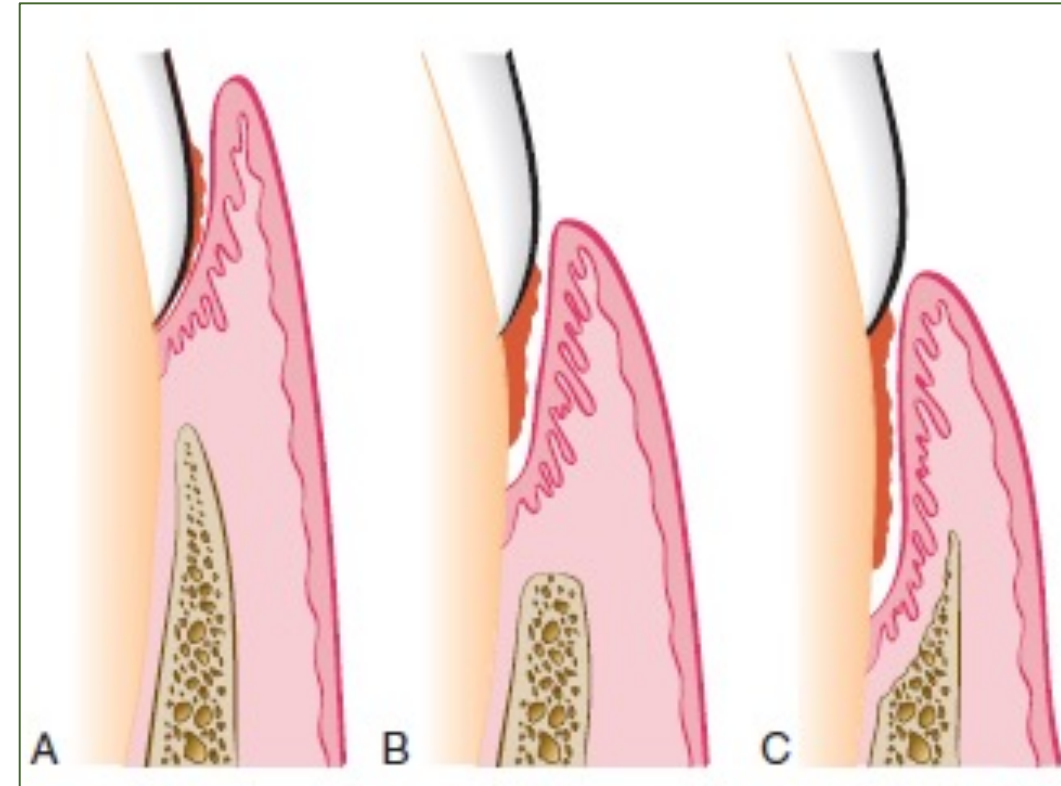


Classification of pockets

1. **Gingival pocket** (“pseudo-pocket”) is formed without destruction of the underlying periodontal tissues
2. **Periodontal pocket** --destruction of the supporting periodontal tissues

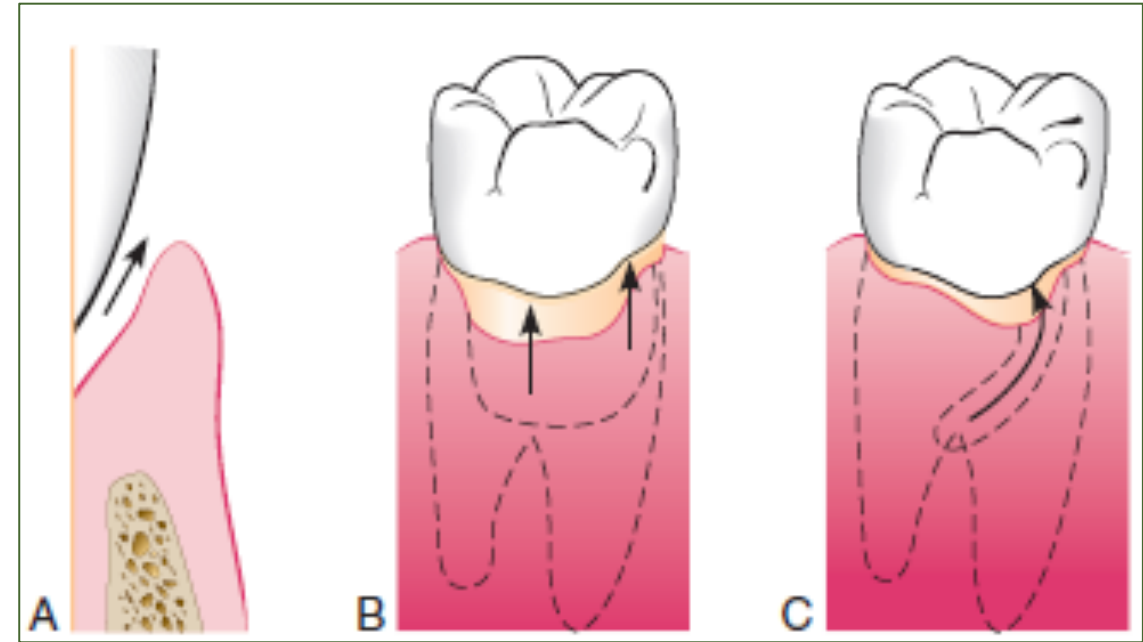
Periodontal pockets can be classified into:

- a. **Suprabony**--bottom of the pocket is coronal to the underlying alveolar bone
- b. **Intrabony** -- bottom of the pocket is apical to the level of the adjacent alveolar bone



(A) Gingival pocket.
(B) Suprabony
(C) Intrabony pocket.

- **Pockets can involve one, two, or more tooth surfaces, and can be of different depths and types on different surfaces of the same tooth**
- **Pockets can also be spiral (i.e., originating on one tooth surface and twisting around the tooth to involve one or more additional surfaces)**
- **Spiral pockets are most common in furcation areas**



Classification of pockets according to involved tooth surfaces.

(A) Simple pocket

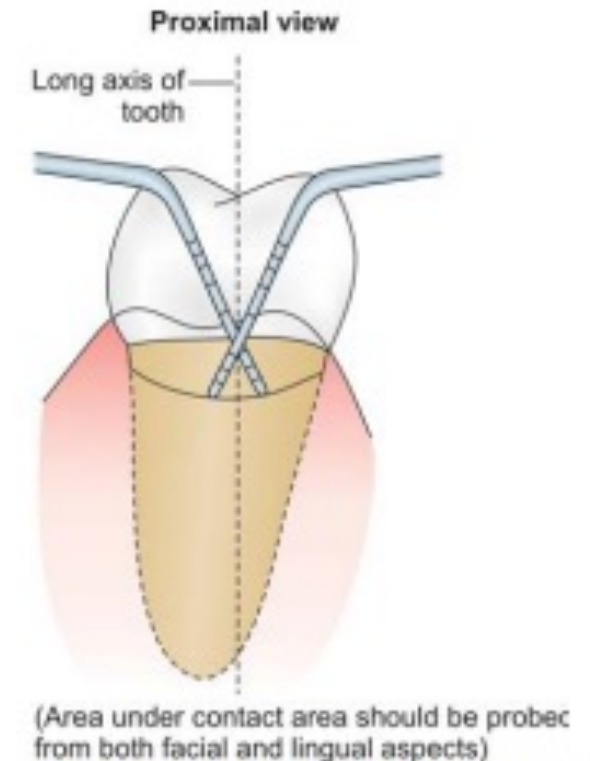
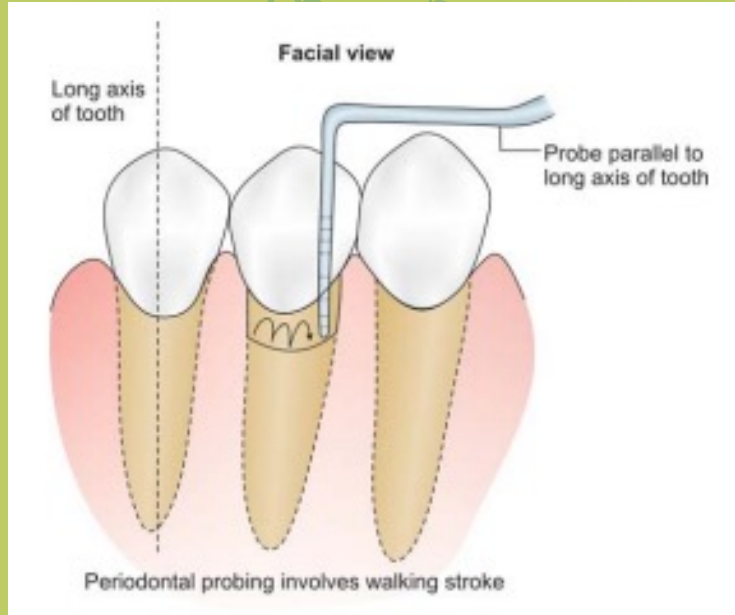
(B) Compound pocket

(C) Complex pocket

Signs and Symptoms of periodontal pockets

- color changes--bluish red marginal gingiva or a bluish red vertical zone that extends from the gingival margin to the attached gingiva
- “rolled” edge separating the gingival margin from the tooth surface
- presence of bleeding, suppuration, and loose teeth
- generally painless
- foul taste in localized areas

The only accurate method of detecting and measuring periodontal pocket is careful exploration with a periodontal probe



Clinical attachment loss [CAL]



In healthy conditions, the connective tissue attachment of the dentogingival junction begins coronally at the CEJ, and the epithelial attachment exists coronal to the connective tissue attachment

- + With attachment loss, CEJ becomes exposed
- + CAL measures the amount of attachment loss that has occurred, with the cementsoenamel junction as the reference point
- + CAL is measured as the distance from the *CEJ to the bottom of the base of the pocket*





Probing depth measurement alone is inadequate for an assessment of periodontitis

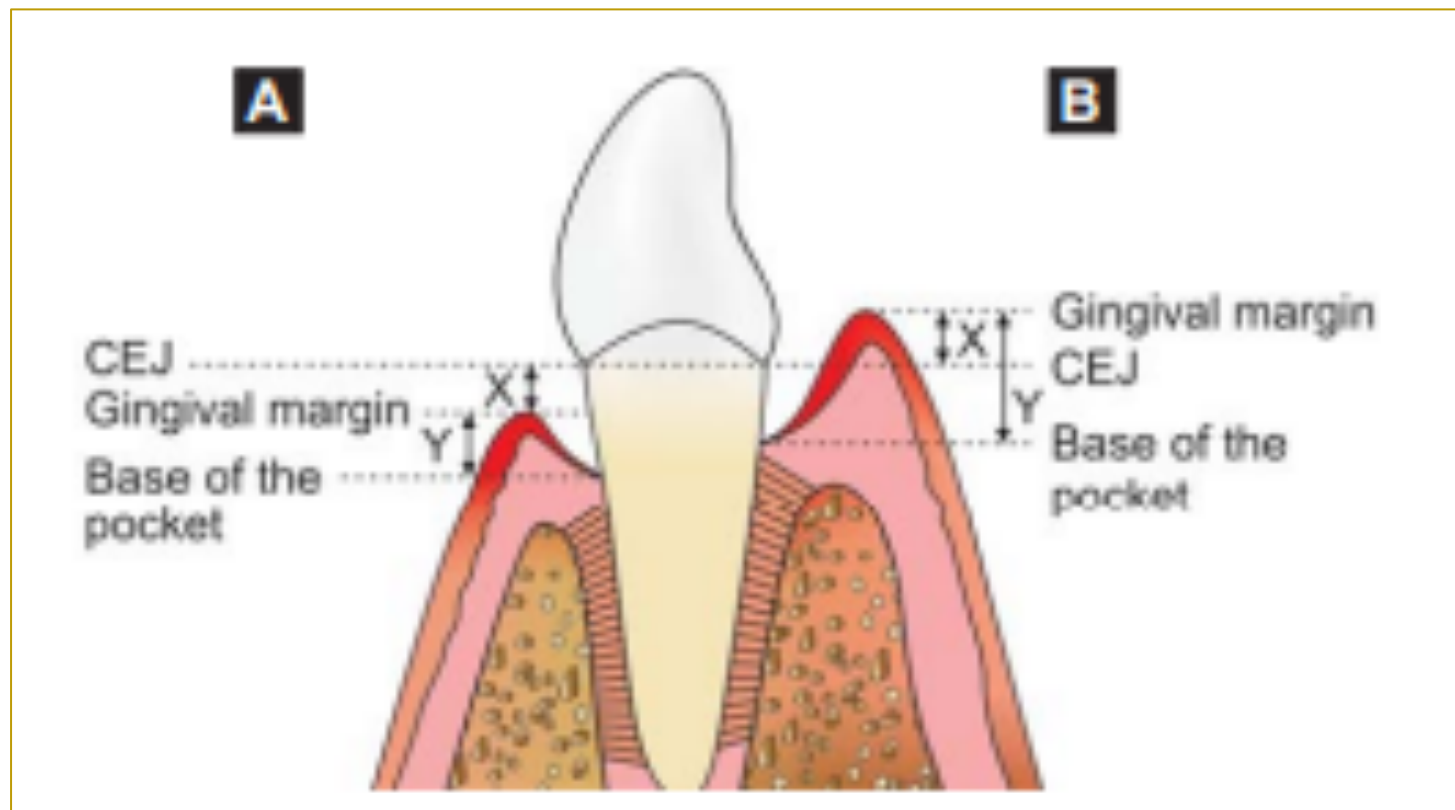


CAL is assessed by adding the extent of gingival recession to the probing depth measurement to estimate the total extent of tissue loss from the CEJ of the tooth



When recession is not visible, height of the marginal gingiva coronal to the CEJ is determined and that measurement subtracted from the probing depth to establish the extent of clinical attachment loss





Calculation of CAL in different situations

- A. When gingival margin is below CEJ -- CAL is calculated by **ADDING** the probing depth (Y) to the distance between CEJ and gingival margin (X) $X+Y$
- B. When gingival margin cover the CEJ, locating on anatomic crown-- CAL is calculated by **SUBTRACTING** the the distance between CEJ and gingival margin (X) from the probing depth (Y) $Y-X$

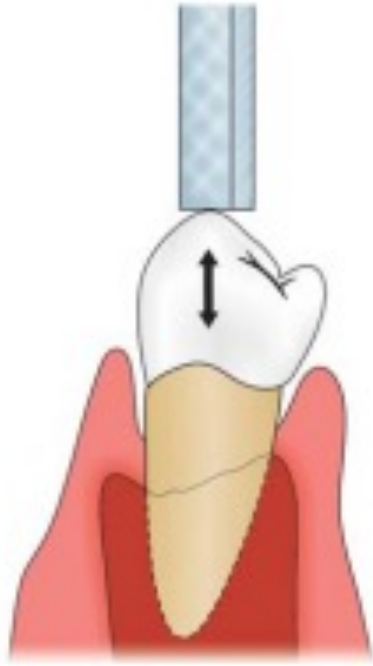
Furcation involvement

- Attachment loss can result in furcation invasion, the pathologic resorption of interradicular bone within a furcation of a multirooted tooth due to periodontal disease
- Detected by carefully probing the root surface for horizontal concavities with a specialized probe such the Nabers probe

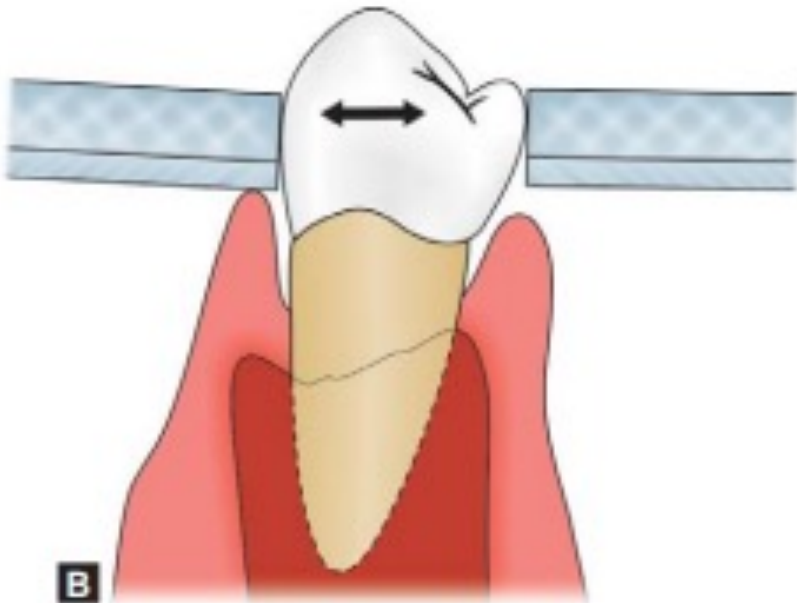


Increased tooth mobility

- **Physiologic mobility is movement up to 0.2 mm horizontally and 0.02 mm axially**
- **Mobility beyond the physiologic range is termed *abnormal* or *pathologic***
- **Single-rooted teeth have more mobility than multi-rooted teeth, with incisors having the most mobility**
- **Three main etiologic factors of tooth mobility are periodontal inflammation, attachment loss, and occlusal trauma**



A



B



C

- A. Tooth mobility checked with one instrument in vertical direction**
- B. Tooth mobility checked with two instruments in buccolingual direction**
- C. Tooth mobility checked with one metal instrument and one finger in buccolingual direction (Not reliable method)**

Change in tooth position-- Pathologic Migration of the Teeth

- **Pathologic migration refers to tooth displacement that results when the balance among the factors that maintain physiologic tooth position is disturbed by periodontal disease**
- **Early sign of disease**
- **Most frequent in the anterior region**
- **Teeth may move in any direction, accompanied by mobility and rotation**
- **Occurs under conditions that weaken the periodontal support, that increase or modify the forces exerted on the teeth, or both**



Labial migration of the maxillary central incisors, especially the right incisor. (A) Frontal view. (B) Lateral view.

Pathologic migration associated with tongue pressure. (A) Facial view. (B) Palatal view.





Radiographic findings of periodontal disease



- Radiographs are valuable for the diagnosis of periodontal disease, estimation of severity, determination of prognosis, and evaluation of treatment outcome
- *Radiographs are an adjunct to the clinical examination, not a substitute for it*
- Radiographs demonstrate changes in calcified tissue



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- + The radiographic survey should consist of a minimum of 14 intraoral films and 4 posterior bitewing films
 - + Panoramic radiographs--simple and convenient, helpful for the detection of developmental anomalies, pathologic lesions of the teeth and jaws, and fractures, as well as dental screening examinations of large groups
 - + They provide an informative overall radiographic picture of the distribution and severity of bone destruction with periodontal disease, but *a complete intraoral series is required for periodontal diagnosis and treatment planning.*



Bone Destruction in Periodontal Disease

- **Early destructive changes of bone that do not remove sufficient mineralized tissue cannot be captured on radiographs**
- **Therefore the earliest signs of periodontal disease must be detected clinically**
- **Radiographs-- indirect method for determining the amount of bone loss -- show the amount of remaining bone**



Patterns of bone destruction

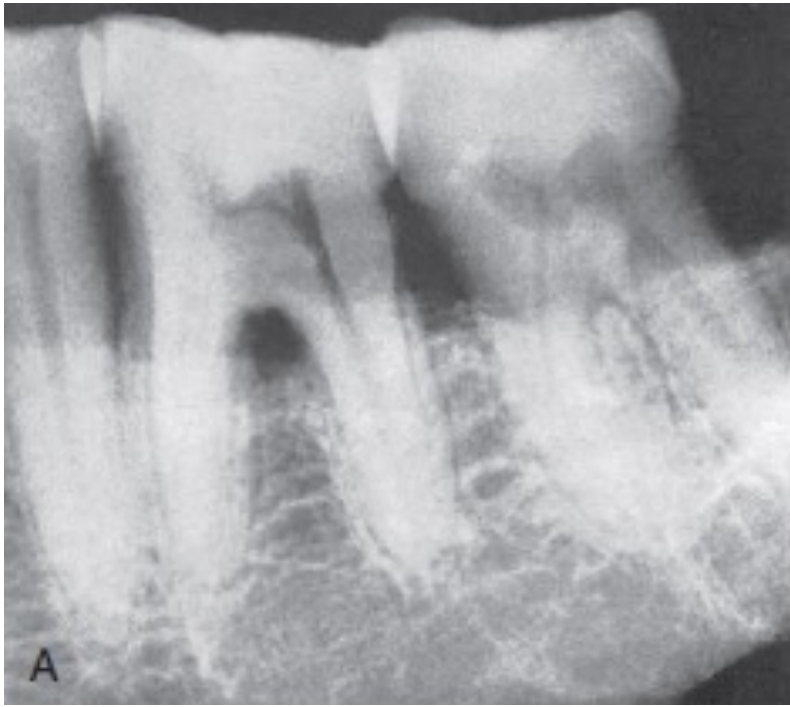
- Horizontal bone loss--height of interdental bone may be reduced, with the crest perpendicular to the long axis of the adjacent teeth
- Angular or Arcuate defects (angular, or vertical, bone loss)



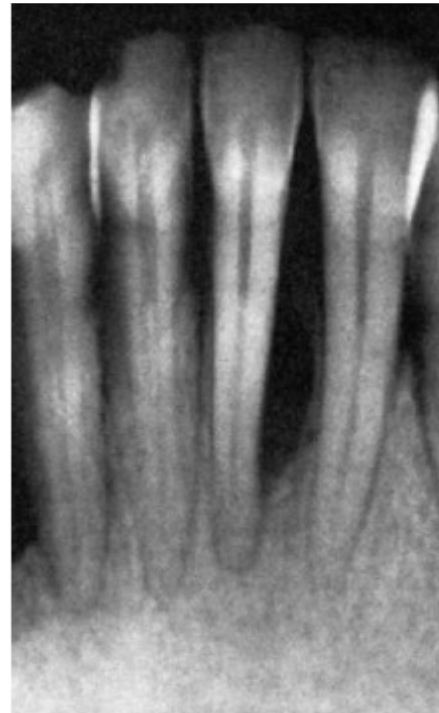
Horizontal bone loss



Vertical bone loss



Furcation involvement



Radiolucent area on lateral aspect of root with chronic periodontal abscess



Widened periodontal space caused by trauma from occlusion



Thank you

